Multi-project scope: Is this tool only for one project at a time, or do you need to manage multiple client projects simultaneously? Your other apps (Archidoc, ArchiSign, Ouvro) — do they already define projects that this should reference?

This will be connected to the main application, which is called Archidoc. You can find a full overview of the Archidoc structure and function below. It is effectively driven by individual projects and the workflow is client project based. As an architectural practice, we can run 15 or 20 projects at once, but it is rare that we need a multiproject context within the scope of an individual project. Having granularity at the level of individual projects is far more important than having aggregated insight, because the workflow is concerned with defining specifications and projects, and managing their execution, which is wholly vertically integrated with individual client projects.

the cross project insight probably is derived from factors such as the firm's capacity or tracking more global insights such as total work, managed, or total fees earned. however, this sort of general accounting intelligence can be derived from our main accounting package which is called Penny Lane. It's important to note that we are not trying to create a full accounting software package — we already have one. However, it lacks the granular flexibility and bespoke possibilities that a dedicated Replit application could offer. This would provide detailed, strict control over the workflow, particularly in terms of its correlation with the nature of the work to be carried out and the quotations and invoices that underpin it. In this context, accounting software is purely about numbers, whereas this new application considers the numbers in relation to the work, including verification, recording, and ensuring that the quotation relates to the specification and is tracked accurately before being entered into the final accounting software.

ARCHIDOC — Architecture & Strategic Overview

ARCHIDOC is a comprehensive technical operating system built for French architectural firms (maîtres d'œuvre). It manages the full lifecycle of construction projects — from initial design and specification through to contractor coordination, site supervision, and document signing. The platform is built on React 19 with TypeScript on the frontend and Express.js on the backend, backed by a PostgreSQL database with over 110 tables managed via Drizzle ORM. Core entities include Projects, Contractors (with multi-contact support and trade classifications), DQE Library Items (detailed quantity estimates following French construction standards), and Construction Tasks. The system already tracks contractor quotations (devis) through a dedicated module where contractors submit pricing against shared DQE specifications, and it maintains formal sign-off workflows between architects and contractors. All data is multi-tenant by user, with session-based authentication, role-based access control, and project-level ownership checks.

For a companion financial tracking application, the key integration points would be ARCHIDOC's existing contractor and project data — specifically the contractor records (company details, contacts, trades), project identifiers, and the quotation/devis data that represents agreed pricing. ARCHIDOC uses a single PostgreSQL database, so the new app could either share the same database with its own tables (simplest path) or run independently and communicate via API. If sharing the database, the new app would need

read access to contractors, projects, and quotation tables while owning its own payment tracking and financial tables. If running as a separate service, it would need authenticated access to ARCHIDOC's existing API layer for contractor and project lookups. Key considerations include: ARCHIDOC uses French construction lot codes and work module classifications that any financial system would need to respect for cost categorisation; all monetary values in quotations are stored as numeric fields; and contractor identification is UUID-based throughout the system. The financial app should treat ARCHIDOC as the source of truth for project scope and contractor relationships, while owning payment schedules, invoice tracking, and financial reporting independently.

Gmail extraction priority: Which documents do contractors typically send you by email — the invoices (situations)? Or do you also receive quotations by email? Understanding the flow helps me design the extraction queue.

All financial information is received by email, either as PDFs or embedded within the email text. Although quotations and invoices are usually in the form of a formal PDF, as they are output from contractors' own accounting software, the AI agent will need to extract data from emails using certain keywords. It will then decide which project the information pertains to, and whether it is relevant. Obviously, keywords such as the client name and address will almost certainly be present. Attachments to these emails can be readily identified once the data has been extracted, since they invariably contain quotation numbers, invoice numbers, and other information that will enable the AI agent to define the nature of the financial information.

I have run tests using Google Notebook LM, and it can easily correlate invoices with original quotations by understanding the content and references within the documents. Clearly, this is an important feature that provides the intelligence required to allocate attachments and understand the financial context within the discipline of the financial workflow.

Feature: This Google Notebook LM feature is extremely useful as it gathers intelligence over time from all the documents related to a single project. It would be interesting to know whether this type of back-end intelligence could also be incorporated into the app, effectively creating a Google Notebook LM-style approach to individual client projects, enabling investigation and questions to be asked in the context of specific client finances. this elevates the app from being a simple financial tracking tool to being a financial advisor and troubleshooter embedded within a client project's finances.

Certificat de Paiement: This seems like a critical output document you generate for clients. Should the app generate these, or is that handled elsewhere (perhaps in Archidoc)?

Yes, the app should manage the payment certificate. A simple integration with Archidoc could provide client information for automatic mailing, etc. However, this is a workflow detail. The important thing is that you understand the context of the certificate of payment within the architectural workflow.

*In the French construction landscape, the "Certificate of Payment"—locally known as a **Certificat de Paiement** or **État d'Acompte**—is a cornerstone of the project management phase (Direction de l'Exécution des Travaux or DET).*

While it might seem like a simple administrative nudge, it serves as a critical bridge between the technical reality of the site and the financial obligations of the client.

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1. The Significance of the Certificate

The certificate acts as a formal validation by the architect that the work claimed by the contractor has actually been performed according to the contract and quality standards.

- ***Verification of Progress:*** *The architect checks the contractor's "situation" (invoice request) against the physical progress on-site.*
- ***Quality Assurance:*** *By signing the certificate, the architect implies that the work being paid for is compliant with the Cahier des Clauses Techniques Particulières (CCTP).*
- ***Financial Control:*** *It accounts for any holdbacks (the retenue de garantie, usually 5%), previous payments, and adjustments for penalties or variations.*
- ***Liability Protection:*** *For the client (the Maître d'Ouvrage), it provides a "buffer." The client is often not an expert; the certificate provides the professional assurance needed to release funds safely.*

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2. Is it a Legal Requirement?

*The answer is a mix of **contractual obligation** and **professional duty**, rather than a singular "statute" in the penal code.*

The Contractual Framework

In France, the relationship is governed by two main frameworks:

1. ***Public Contracts (Marchés Publics):*** *It is strictly a legal requirement under the Code de la commande publique. The architect must certify the "service rendered" before the public accountant can pay.*
2. ***Private Contracts (Marchés Privés):*** *It is technically a **contractual requirement**. However, because almost all private contracts in France use the **NF P 03-001 standard** (the benchmark for French construction contracts), it becomes a binding part of the workflow.*

The Role of the Ordre des Architectes

*The Ordre des Architectes (the French governing body) embeds this responsibility into the **Code of Professional Duties**.*

- ***Deontological Obligation:*** *Architects are ethically bound to monitor the financial execution of the contract. Failing to issue certificates or doing so negligently can lead to disciplinary action from the Order or professional indemnity claims.*

- ***The "Duty of Advice":*** *Under French law, the architect has a permanent devoir de conseil. If an architect allows a client to overpay a contractor for work not yet done, the architect can be held legally liable if the contractor goes bankrupt.*

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3. The Standard Workflow

The process follows a rhythmic, monthly cycle defined by the standard phases of a French project:

Step	***Participant***	***Action***
1. Draft	*Contractor*	*Submits a Projet de Décompte (Draft Bill) based on work done.*
2. Review	*Architect*	*Verifies the site, applies the "formula" (deducting holdbacks/advances).*
3. Issuance	*Architect*	*Issues the* ***Certificat de Paiement*** *to the Client.*
4. Payment	*Client*	*Pays the contractor based strictly on the architect's certified amount.*

Summary

The certificate is not just a polite suggestion; it is the ***legal shield for the client*** *and a* ***professional mandate for the architect****. In France, a client who pays a contractor without an architect's certificate is essentially "flying blind" and loses significant legal leverage should a dispute arise regarding the quality or completion of those works.*

Fee tracking (RS1/RS2/AGENT): Your fee calculations are a significant part of this spreadsheet. Should the financial app also track your own invoicing to the client, or just the contractor side?

Yes, fee calculations are critical, and the app should track fees. To this end, we will need to develop your understanding of how our fees are charged, because there are basically three types of fee charge.

1. *Fees derived as a percentage of the net TVA cost of the Works, this is always a percentage of the Works and is never a fixed fee*

2. *There is a separate fee associated with the design and planning permission process, which precedes the construction project. This fee is always fixed and agreed with the client. In this respect, it does not appear in the financial tracking workflow of the live project; instead, it is recorded on a separate tab in our Excel spreadsheet, alongside the other financial information. In reality, however, this fee is a prerequisite to construction and therefore needs to be handled in a separate interface within the application.*
3. *A third potential fee applies to projects involving only the planning application, or where the planning application fee is quoted separately from the conception fee. In these cases, it is an individual, one-off fee, negotiated and agreed with clients at the outset. Like the conception fee, it is not embedded in the workflow of the live project, but precedes it.*

Once these fees have been set, we will generate invoices from our invoicing system, which is currently located at https://www.pennylane.com/fr. We should be able to integrate this software in the future if required. However, due to the volume of our invoicing, it is probably better to maintain a manual bridge between the Replit app and our accounting software, simply carrying over the figures. In this respect, our main accounting system is unaware of the integrity of the financial workflow, which must be managed inside the new Replit application.

The PV/MV (variations) workflow: How important is it to track variations formally? I see the PVMV tab has a structured approach with plan references and outcomes.

*In the French construction industry, the terms **Plus-value (PV)** and **Moins-value (MV)** are essential for managing the financial gap between the initial contract and the final reality of a building project.*

*Think of them as **financial adjustment bookmarks.***

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1. What are they?

In a standard French construction contract (often a marché à prix global et forfaitaire), the price is set at the start. However, changes are inevitable.

- ***Plus-value (PV):** An **increase** in cost. This happens when the client asks for a more expensive material, adds a new room, or if a technical necessity arises that wasn't in the original plan.*
- ***Moins-value (MV):** A **reduction** in cost. This occurs when a specific task is canceled, a cheaper material is substituted, or the quantity of work actually performed is less than what was quoted.*

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2. The Financial Workflow

For your application, the agent needs to see this as a dynamic balancing act. Here is how it typically flows:

Step A: The Variation Order (Avenant)

*Before any PV or MV is officially recorded in the accounting books, the contractor and the client usually sign an **Avenant** (an amendment). This document proves that both parties agree to the change in price.*

Step B: The Monthly Statement (Situation de Travaux)

French contractors don't just send one big bill at the end. They send monthly "Situations."

- *The **PV** is added to the total work completed.*
- *The **MV** is subtracted from the total.*
- *The agent sees a "Net" amount to be paid for that period.*

Step C: The Final Account (Décompte Général Définitif - DGD)

At the end of the project, all the PVs and MVs are tallied up.

- ***Final Price** = Original Contract Price + (Sum of all PV) - (Sum of all MV).*

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3. Why it Matters for the Agent

If the agent is reviewing a loan or a grant application, they need to understand that:

1. ***PVs are not "hidden fees":** They represent extra value or necessary changes.*
2. ***MVs are "credits":** They ensure the client isn't paying for work that wasn't done.*
3. ***The Budget is Fluid:** A project starting at €200,000 might end at €210,000 because the PVs outweighed the MVs.*

> ***Key Takeaway:** In French accounting, these terms ensure the "True Cost" of the construction is reflected, rather than sticking blindly to an outdated initial quote.*

Another aspect to understand in terms of the administrative framework and context for this tracking app is the Marché de Travaux Privé. We would expect this to be managed and created inside the app, since it is integral to financial workflow management.

*In the French private sector, the **Marché de Travaux Privé** (Private Works Contract) is the "Source of Truth." For an application managing quotes and invoices, this document is the foundation upon which every digital transaction is built.*

Without a signed Marché, a quote is just a proposal; with it, the quote becomes a legally binding financial framework.

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1. The Financial "North Star"

*In a private project (governed by the Code Civil and often the **NF P 03-001** standard), the contract defines how money will move. It moves the project from a "shopping list" phase into a "legal obligation" phase.*

Key Financial Pillars:

- ***The Price Type:** It defines if the project is Prix Forfaitaire (Fixed Total Price) or Prix Unitaire (Unit Price). This tells your application whether to flag a change as an "error" or a "valid adjustment."*
- ***Payment Schedule (Échéancier):** It dictates the triggers for invoices (e.g., 20% at foundation, 30% at "Out of Water" status).*
- ***The Holdback (Retenue de Garantie):** In France, it is standard to withhold **5%** of each invoice to ensure the work is finished correctly. Your app must track this "virtual" money that isn't paid until a year later.*

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2. The Documentation Landscape

The Marché is not just one piece of paper; it is a "bundle" of documents. If an agent sees an invoice that doesn't match the quote, they refer to this hierarchy:

Document	***Financial Role***
L'Acte d'Engagement	*The "Marriage Certificate." It states the final agreed price and who is paying whom.*
Le CCAP	*The "Rulebook." It defines penalties for delays and how Plus-values are handled.*

Le CCTP	*The "Technical Specs." This justifies the costs (e.g., why the tile costs €80/m² instead of €20/m²).*
Le Planning	*The "Clock." It links time to money, allowing the agent to see if the invoicing speed matches the build speed.*

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3. Role in the Quotation & Invoicing Workflow

*For your application's logic, the Marché Privé acts as the **Validation Gate**:*

1. ***Baseline:*** *The app imports the Marché total as the "Initial Budget."*
2. ***Quotations (Devis):*** *Any new quote generated mid-project is checked against the Marché. If it changes the scope, it must be converted into an **Avenant** (Amendment) to update the contract value.*
3. ***Invoicing (Situations):*** *The app generates invoices based on the percentage of progress relative to the Marché's original breakdown.*

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4. Why the Agent Needs to Know This

*If an agent is looking at an invoice for €10,000, but the original quote was for €8,000, the Marché Privé is the only reason they can legally pay it. It provides the **legal authority** for the financial adjustments (the Plus-values we discussed earlier).*

Without understanding the Marché, the agent is just looking at numbers in a vacuum. With it, they are managing a strictly regulated financial flow.

I want to stay in planning mode until we thoroughly understand the context, shape and architecture for this application. we need to be clear about, the workflow documents and expectation of the user in terms of how the application will manage the financial data. in this briefing document, I've tried to explain that this needs to be more than a simple tracking tool. We want to leverage the agentic approach and inherent intelligence such as the Google Notebook LM, model, to leverage true intelligence and insights to help the user, manage the finances with the minimum of intervention. in this respect. Picking up on errors using intelligence to identify. embedded errors, miscalculations and or miscorrelation between. quotations and invoices. here is a list of the common recurring issues we have to deal with that should be considered in the overall architecture..

Modes of invoicing

For your agent to manage an architectural firm's finances in France, it needs to recognize that these aren't just "different styles"—they are two distinct financial workflows dictated by the scale and the legal framework of the project.

*In the French market, these are generally categorized as **Direct Billing** (Simple) and **Progress Billing** (Complex).*

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1. Mode A: The "Devis/Facture" (Direct Billing)

Common with artisans and small enterprises (SMEs) on smaller private renovations.

- ***The Logic:** The quotation (Devis) is a fixed list of tasks. When a task is done, it is invoiced.*

***The Workflow:** 1. **Devis:** €10,000 total.*

*2. **Acompte:** A deposit invoice (usually 30%) is issued to start.*

*3. **Facture Intermédiaire:** An invoice for a specific completed item (e.g., "Installation of windows").*

- *4. **Facture Finale:** The balance remaining.*
- ***The Challenge for the Agent:** It is easy to "count to zero," but it doesn't account for work that is "half-done." It's binary: either it's invoiced or it isn't.*

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2. Mode B: The "Situation de Travaux" (Progress Billing)

*Standard for mid-to-large companies and any project following the **NF P 03-001** standard. This is what you find difficult to track.*

- ***The Logic:** You never invoice an "item"; you invoice the **evolution** of the entire project. The document is a spreadsheet that stays alive from day one until the end.*
- ***The Workflow:** Every month, the contractor submits a Situation (Statement). It lists every line item from the original quote and adds a column for **"% Complete."***

- *Example: If the masonry is 50% done, they invoice 50% of that line item.*

The Calculation: *The invoice amount is:*

-
- $$(\text{Cumulative Work Done}) - (\text{Total Previously Invoiced}) = \text{Net to Pay this Month}$$

Why this is hard to "Count to Zero":

1. ***Cumulative Nature:*** *Each invoice includes the totals of all previous invoices. If the agent adds them all up manually, they will accidentally triple-count the money.*
2. ***The "Retenue de Garantie":*** *As mentioned before, 5% is often sliced off the top of every Situation, making the "Net to Pay" lower than the "Work Value."*
3. ***The Revision of Prices:*** *In many French contracts, the price can be adjusted monthly based on inflation indexes (the BT01 index). This means the "Zero" point is actually a moving target.*

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3. Comparison for the Financial Agent

Feature	*Devis/Facture (Simple)*	*Situation de Travaux (Complex)*
Document Type	*Static Invoice*	*Cumulative Statement*
Tracking	*Subtracting invoices from a total*	*Tracking % of completion per line*
Best For	*Kitchen fit-outs, small repairs*	*New builds, major structural work*

Risk	*Paying for work not yet finished*	*Mathematical errors in cumulative totals*

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4. How the Application Should Handle This

*To give your agent "absolute clarity," the application must treat the **Initial Quotation (Marché)** as the master template.*

- ***For Mode A:*** *The app simply ticks off lines as they are billed.*
- ***For Mode B:*** *The app must store the **Cumulative Total**. Instead of looking at individual invoices as isolated events, it should display a **Progress Bar** for each line item of the quote.*

> ***Pro Tip for the Agent:*** *In a Situation de Travaux, the only number that matters for "counting to zero" is the **"Reste à Réaliser"** (Remainder to be done) column. If your app extracts that specific data point, the complexity disappears.*

*In the French architectural workflow, the **Certificat de Paiement** (often called the Proposition de Paiement) is the "Filter" through which all contractor invoices must pass before the client pays a single cent.*

*For your application's agent, this is the most critical document because it represents the **Architect's professional stamp of approval** on the financial data.*

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1. The Architect as the "Financial Referee"

*A contractor (especially in **Mode B/Progress Billing**) will often claim they have finished 80% of a task to get paid faster. The Architect's job is to visit the site, look at the work, and say: "No, it's only 60%."*

*The **Certificat de Paiement** is the document where the Architect "corrects" the contractor's math.*

The Workflow:

1. ***Contractor sends a "Project de Décompte":*** *This is their draft invoice based on their view of progress.*

2. ***Architect Reviews:*** *The Architect checks the site and the contract.*
3. ***Architect issues the "Certificat de Paiement":*** *This is a formal document sent to the Client. It says: "We have verified the work; you should pay the contractor €X amount."*
4. ***Client Pays:*** *The client (or their bank) only releases funds based on the Architect's certificate, not the contractor's invoice.*

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2. What's inside the Certificate?

For your agent to track this, the certificate acts as a ***reconciliation statement****. It usually breaks down the math like this:*

Component	***Calculation Logic***
Montant Travaux HT	*Total value of work done since the start of the project.*
Plus-values / Moins-values	*Adjustments for changes (as we discussed earlier).*
Paiements Antérieurs	*Subtraction of everything already paid to date.*
Retenue de Garantie (5%)	*Deduction of the "safety" holdback.*
Net à Payer (TTC)	*The final, actual amount the client must wire to the bank.*

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3. Integration into your Application

If your application is managing quotes and invoices, the ***Certificat de Paiement*** *is the "Final Record."*

- ***The Problem:*** *A contractor might send an invoice for €12,000.*
- ***The Reality:*** *The Architect certifies only €10,000.*
- ***The App's Logic:*** *Your application should not mark the "Initial Quote" as having €12,000 deducted. It must follow the* ***Certified Amount****.*

Agent Insight: *In French law, the Architect is "Maître d'Œuvre" (Master of Works). Their certificate protects the client from overpaying and protects the contractor from being ignored. It is the legal bridge between the* ***Construction Site*** *and the* ***Bank Account****.*

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4. Why this Solves the "Counting to Zero" Problem

The reason it's hard to count to zero with just invoices is that contractors often make mistakes or "pad" their numbers.

By using the ***Certificat de Paiement*** *as the data source:*

- *The agent sees the* ***official progress*** *(e.g., Project is 42% complete).*
- *The agent sees the* ***actual remaining budget*** *(Reste à payer).*
- *The agent has a paper trail if the client asks: "Why is there still €50,000 left on the quote if the house looks finished?" (The answer is usually found in the Retenue de Garantie or unapproved Plus-values).*

To help your agent visualize how a single construction task moves from a simple quote to a final payment in the French system, here is the "Data Map" of a typical workflow.

The Journey of a Single Line Item

Imagine a quote for ***"Stone Masonry"*** *originally priced at* ***€10,000****.*

Phase	***Document***	***Action/Calculation***	***Resulting Value***
1. The Origin	***Devis (Quote)***	*The contractor sets the initial price.*	***€10,000***

2. The Change	***Avenant (PV)***	*Client requests premium stone. A **Plus-value** is added.*	***€12,000** (New Total)*
3. The Claim	***Situation (Progress)***	*Contractor claims the wall is **75%** built.*	***€9,000** (Work Value)*
4. The Audit	***Architect's Visit***	*Architect sees it is only **50%** built.*	***€6,000** (Validated Value)*
5. The Security	***Retenue de Garantie***	*The Architect deducts **5%** for the legal holdback.*	***€5,700** (Net HT)*
6. The Payment	***Certificat de Paiement***	*Architect adds VAT (20%) and issues the cert.*	***€6,840** (Total TTC to pay)*

How the Application Tracks This

For the agent to "count to zero" with absolute clarity, the application must manage three distinct "buckets" of data for every line item:

1. ***The Contracted Bucket (The "Goal"):** This is the **Initial Quote + Avenants (PV/MV)**.*
    - *Example: €12,000.*
2. ***The Certified Bucket (The "Reality"):** This is the total cumulative work validated by the Architect across all previous certificates.*
    - *Example: €6,000.*
3. ***The "Reste à Réaliser" (The "Remaining"):** This is the simple subtraction of Bucket 1 minus Bucket 2.*
    - *Example: **€6,000 left to build.***

Why the "Situation" Style is Actually Better (Once Mastered)

*While the **Mode B (Progress Billing)** feels more complex than a simple invoice, it provides the agent with a **Financial Health Check** that simple invoicing cannot:*

- ***In Mode A (Simple):*** *You only know what has been paid. You don't know if the contractor is ahead or behind schedule.*
- ***In Mode B (Progress):*** *Because the Situation shows a percentage against every single line item of the Marché, the agent can see exactly where the project is stalling.*
    - *Example: If "Roofing" is at 0% but "Interior Painting" is at 50%, the agent (and the Architect) can flag a sequence error immediately.*

Final Summary for the Agent

In the French private construction landscape, the ***Certificat de Paiement*** *is the ultimate truth. It reconciles the contractor's ambition (the Situation) with the physical reality of the site (the Architect's verification), all while respecting the legal framework of the* ***Marché Privé****.*